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Nonfluorescent Molecule Detection in 10^2 nm Nanofluidic Channels by Photothermal Optical Diffraction

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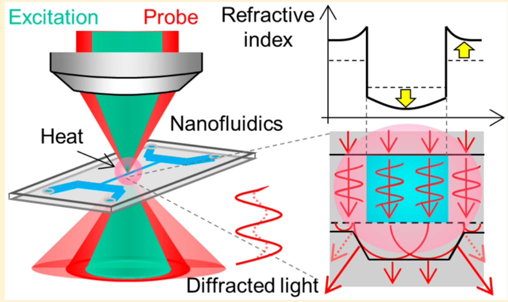
ABSTRACT: Integrating analytical systems in 10^1−10^3 nm spaces provides ultrasensitive analytical devices at the single cell and the single molecule levels due to the ultrasmall space, and fundamental technologies for nanofluidics are developed. A simple and ultrasensitive detection method is one of the essential technologies for nanofluidics; however, it is still challenging due to the ultrasmall volume at the attoliter to femtoliter scale. In this study, we report a new photothermal detection method of nonfluorescent molecules for a 10^2 nm space, photothermal optical diffraction (POD), which utilizes light absorption and heat generation by an analyte and optical diffraction by a nanochannel after heat diffusion. Concentration determination of nonfluorescent molecules in a 400 nm channel was successfully demonstrated, and a limit of detection (LOD) of 5.0 μM was achieved, corresponding to 500 molecules (0.84 zmol) in a detection volume of 230 aL. Also, detection in a 200 nm channel was successfully demonstrated without degradation of the LOD. Our method can be widely used for chemical and biological analyses in 10^2−10^3 nm nanofluidics.



Microfluidics is progressing rapidly in the past few decades, where chemical and biological processes are integrated on a centimeter-scale substrate with fabricated microfluidic channels.[1−5] Recently, the integration is further progressing to 10^1−10^3 nm spaces (nanofluidics),[6,7] which enables handling of an ultrasmall sample and high-efficiency processing at the single-molecule level. For example, size separation of DNA fragments by a nanopillar channel[8] and a chromatographic separation by the large surface-to-volume ratio of nanochannels[9] have been realized. Almost 100% efficient immunoreaction[10] and high-throughput DNA sequencing[11] with a nanopore have also been developed. From the viewpoint of fundamental research, nanochannels and nanopores are used as an artificial ion channel in a lipid bilayer to study transport properties of different ions or molecules.[12,13] Sample injection to living cells and sampling from a living single-cell have also been realized.[14,15] In addition, unique fluid properties at the 10^1−10^2 nm scale have been reported: ionic mobility and distribution,[16,17] solute and solvent transport,[18] hydrodynamic flow profile,[19] and structure of water.[20] Device applications utilizing these unique properties have been developed, such as energy conversion,[21,22] chiral recognition,[23] and water desalination.[24]

As described above, 10^1−10^3 nm spaces are now gathering much attention in fundamental sciences and device applications. However, molecular detection in 10^1−10^3 nm spaces is challenging due to the ultrasmall space. For example, a 1 nM solution in a 1 fL detection volume corresponds to less than a single molecule. Therefore, an ultrasensitive molecule detection method is required for nanofluidic devices. Label-free detection is also essential for targeting a wide range of molecules. Laser-induced fluorescence (LIF) is now widely used due to the high sensitivity,[25−27] while analytes with native fluorescence are limited and the LIF usually require labeling processes. Several papers reported label-free detection methods for nanofluidics. For example, electrical/electrochemical detection methods were used, and single-molecule detection was reported by a solid-state nanopore[28] and redox cycling in a nanogap transducer.[29] Resonance-based detection methods such as surface plasmon resonance (SPR)[30] and micro- or nanocavity[31] are also widely known as sensitive label-free molecule detection methods, but these methods require complicated fabrication of nanostructures. A simple detection method without additional nanofabrication is desired.

Photothermal spectroscopy (PTS) is well-known as an ultrasensitive detection method for nonfluorescent molecules. The principle of the PTS is based on light absorption and heat generation by target molecules. This effect is common to almost all molecules, thus the PTS has wide applicability. Among the PTS, thermal lens spectrometry (TLS) has been widely used because of the high sensitivity. The TLS was initially applied to millimeter−centimeter scale analytical tools.[32] For microfluidics, TLS was realized under a microscope: a thermal lens microscope (TLM).[33−35] A limit of detection (LOD) of 0.4 molecules was demonstrated in a microfluidic channel (a detection volume of 7 fL).[36] Furthermore, microchannel-assisted thermal lens spectroscopy (MATLS) was developed as a wavelength-tunable TLM.[37] MATLS utilized rapid thermal conduction of a glass substrate and realized thermal lens detection in a microchannel without

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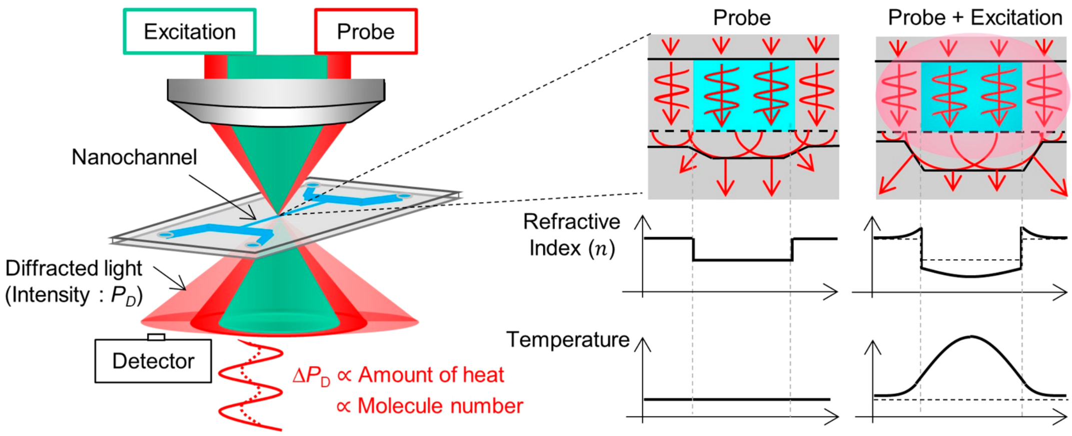

Figure 1. Principle of the POD.

focusing an excitation beam. For a nanofluidic channel, however, it is difficult to apply the TLM. The TLM is based on geometric optics (a lensing effect) and cannot be applied to the 10^1–10^2 nm channels which are smaller than a wavelength of visible light. To overcome this, a differential interference contrast-thermal lens microscope (DIC-TLM)[38] was developed by introduction of optical interference to the TLM. The principle of the DIC-TLM is based on wave optics (an interference effect) and thus applicable to nanochannels. By utilizing the DIC-TLM, a label-free protein molecule detection in a 21 μm × 900 nm channel has been achieved with the LOD of 600 molecules.[39] Also, it was applied to an enzyme-linked immunosorbent assay (ELISA) in a 700 nm channel and realized the detection of a countable-number of protein molecules.[40] However, for sizes smaller than 900 nm, the sensitivity of the DIC-TLM rapidly decreased due to thermal diffusion to the glass substrates.[39] For 10^1–10^2 nm channels, the effect of thermal diffusion on the glass substrates becomes dominant. For example, the thermal diffusion length at 1 kHz modulation is calculated to be ~7 μm in water, which is one order larger than the 10^2 nm nanochannels. Then, most of the heat generated by the photothermal effect does not contribute to the signal, and the detection performance deteriorates. This limitation hinders the DIC-TLM from applying the 10^1–10^2 nm channel. For single-molecule processing, smaller channels are necessary to provide the precise molecular processing. For this reason, a novel detection principle which can overcome this limitation is required.

In this study, we overcome this limitation by introducing a new detection principle: photothermal optical diffraction (POD), which utilizes an enhancement of optical diffraction from a nanochannel after light absorption, heat generation, heat diffusion, and refractive index change. As shown later, the heat diffusion to the substrate also contributes to the POD signal, which enables a sensitive nonfluorescent molecule detection in 10^2 nm channels. No additional nanostructures are required for detection, and the POD is widely applicable to the nanochannels.

EXPERIMENTAL SECTION

Principle of the POD Detection Method. Figure 1 shows the principle of the POD. Because the width of the nanochannel (10^1–10^2 nm) is smaller than the focused probe beam spot (~micrometers), a part of the probe beam passes through the glass substrate. The speed of light is dependent on the refractive index of the propagating media; therefore, the refractive index difference (Δn) between water (n = 1.33) and glass (n = 1.46) gives rise to a phase difference between the probe beams passing through the channel and the substrate. This phase difference induces a wavefront distortion at the water/glass interface, which is observed as optical diffraction in the far-field. The extent of optical diffraction depends on the refractive index difference between glass and water (Δn). The intensity-modulated excitation beam (CW mode) is focused and absorbed by the target nonfluorescent molecules in the nanochannel, followed by a nonradiative thermal relaxation and a localized temperature change of the water and glass phases. This temperature change (ΔT) induces refractive index changes of the water and glass phases because the thermal diffusion length (~micrometers for 1 kHz excitation beam modulation) is larger than channel size (10^1–10^2 nm). The dn/dT of glass and water are 9.8×10^{-6} and -9.1×10^{-5}.[41] Therefore, the Δn becomes larger by the photothermal effect, which enhances the diffracted light intensity, which is detected by a photodetector. The diffracted light intensity change is measured as a signal, which is proportional to the number of target molecules in the nanochannel. The principle of the POD utilizes the optical diffraction by a single nanochannel. Due to the small size of the nanochannel, the diffraction angle becomes large, which enables a separation of the transmitted light and the diffracted light, leading to the detection of only the diffracted light.

The important feature of the POD is that it is possible to use heat diffusing to the glass substrate. In the DIC-TLM, the heat diffusion to the glass substrate leads to a thermal loss and a cancelation effect (dn/dT, positive for glass and negative for water) which decreases the sensitivity of the DIC-TLM and make it difficult to apply to ultrasmall channels with a size less than 900 nm.[39] In the POD, however, the heat diffusion to the glass substrate increases Δn, leading to an increase of intensity of optical diffraction. The heat diffusion to the glass substrate is inevitable in the ultrasmall space. Thus, the principle of the POD is essential for an ultrasensitive detection in nanochannels.

Optical Systems. Figure 2a shows an optical system of the POD. A 532 nm solid-state laser (Verdi-5M, Coherent Inc.) was applied as an excitation beam. The intensity of the excitation beam was modulated by a light chopper (5584A, NF Corporation, Japan) at a modulation frequency of 1.1 kHz. A 633 nm He–Ne laser (25-LHP-925, Melles Griot Inc.) was

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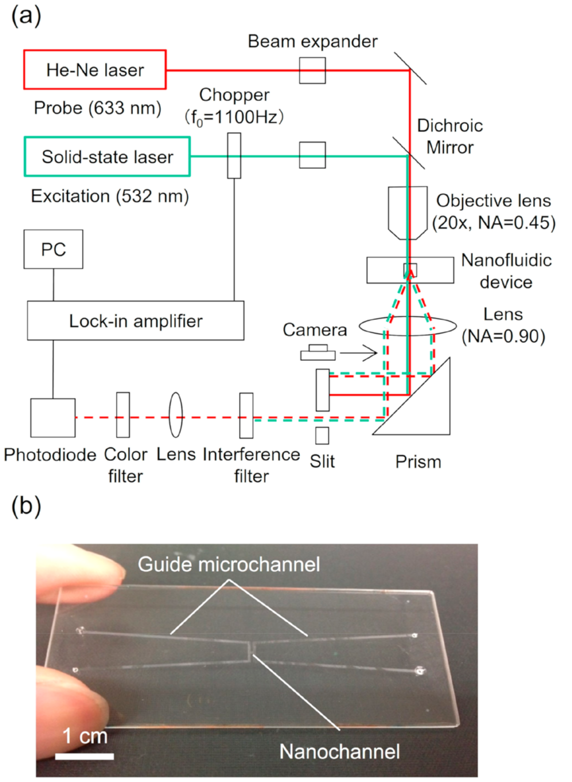

Figure 2. Illustration of the POD system: (a) an optical system of the POD (dashed lines indicate a diffracted light by a channel) and (b) a photograph of the nanofluidic device.

applied as a probe beam. Both beams were focused on a nanochannel through an objective lens (magnification: 20×; NA, 0.45) attached to a microscope (BX51, Olympus Corporation, Japan). The power of the excitation and the probe beams were 50 mW and 5 mW under the objective lens, respectively. Twelve nanochannels with 50 μm pitch were fabricated on a fused-silica glass substrate (70 mm × 30 mm, VIOSIL-SQ, Shin-Etsu Chemical Co., Ltd., Japan) using electron-beam lithography and reactive-ion etching. The length of the nanochannels is 400 μm, and the width and depth were 400 nm unless otherwise mentioned. Guide microchannels with an inlet hole were fabricated on another glass substrate by photolithography and reactive-ion etching. The width and depth of the microchannels were 500 μm and 5 mm, respectively. Microchannels and nanochannels were connected by a thermal bonding method. The detailed fabrication process was described elsewhere.[42] The fabricated device is shown in Figure 2b. A sample solution was injected into nanochannels via microchannels by applying pressure to the inlet hole. The nanofluidic device was fixed on a 2-D stage (BIOS-235S, Sigma Koki Co., Ltd., Japan) which had 100 nm steps for each direction and allowed for adjustment of the channel position precisely. The probe beam spot diameter was approximately 1.7 μm, which was almost 6-times larger than the width of the nanochannel. Therefore, a part of the focused beams was diffracted by the single nanochannel. After passing through the channel, the probe and excitation beams were collected and converted into parallel lights by a collection lens (NA = 0.90), and the diffracted light was separated from the transmitted light with a slit (4 mm width and 10 mm height). Finally, the excitation beam was cut by an interference filter and a color filter, and the remaining diffracted probe beam was focused and detected by a photodiode with a 400 μm aperture pinhole (ET-2030, Electro-Optics Technology Inc.). The detected electric signal was fed into a lock-in amplifier (LI5660, NF Corporation, Japan), and the component synchronized to the chopper frequency was extracted to selectively detect the probe beam intensity change by the photothermal effect of target molecules in the channel. The time constant of the lock-in amplifier was set to 1 s. For comparison of the performance, a DIC-TLM was utilized. The wavelength and power of the excitation and probe beam were the same as those of the POD under the objective lens (NA = 0.75).

RESULTS AND DISCUSSION

Principle Verification. First, optical diffraction by a nanochannel was confirmed. Figure 3 shows images of the probe beam spot after passing through the collection lens.

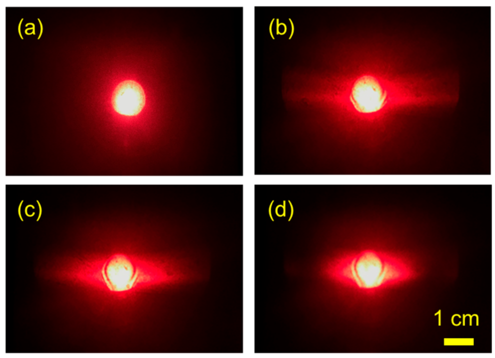

Figure 3. Typical images of optical diffraction by a single nanochannel: (a) without a channel and (b−d) with a channel having a depth of 300 nm depth and widths of (b) 300 nm, (c) 500 nm, and (d) 700 nm.

Images were taken by a camera set at the position as shown in Figure 2. In this experiment, only a probe beam was focused on the nanochannel by an objective lens (NA = 0.30). The channel width was changed from 300 to 1100 nm, and the channel depth was fixed at 300 nm. The channel was empty (filled with air). As shown in Figure 3, the diffracted light was observed in the direction perpendicular to the nanochannel. This anisotropy of the diffracted light can be explained as follows. The channel width was narrower than the diameter of the probe beam spot, and the probe beam was diffracted based on the mechanism as mentioned above. On the other hand, the channel length was longer than the diameter of the probe beam spot, and there was no phase difference in the longitudinal direction of the nanochannel. Therefore, the probe beam was not diffracted in the longitudinal direction. Also, it was confirmed that the diffraction angle became smaller as the channel width became wider, which was well consistent with the Fresnel−Kirchhoff diffraction theory.[43] This result indicates that a smaller channel is suitable for the measurement of optical diffraction because a large diffraction angle enables easy separation of the diffracted light from the transmitted light. Previously, several label-free detection methods using multiple nanochannels as a diffraction grating were reported.[44,45] The diffraction angle is fixed by a spacing of the channels and independent of Δn. Then, it is difficult to measure the diffraction angle change due to the light absorption and heat generation. In this way, single-channel diffraction allows for sensitive detection of Δn via a change of

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the refractive index and the diffraction angle arising from light absorption of the analyte.

Next, the diffracted light intensity change by the photothermal effect was confirmed. To investigate the dependence of the signal on optical diffraction, the detection position by the slit was scanned from the center region of the transmitted light to the diffracted light region. For a precise position measurement, the size of the probe beam was expanded to 9 mm in radius at the slit position using lenses, and the slit width was set to 1 mm in this experiment. An aqueous solution of a nonfluorescent dye, Sunset Yellow FCF (Wako Pure Chemical Corporation, Osaka, Japan), was used as a sample. The absorption coefficient is 4800 M^{-1} cm^{-1} at 532 nm. Figure 4

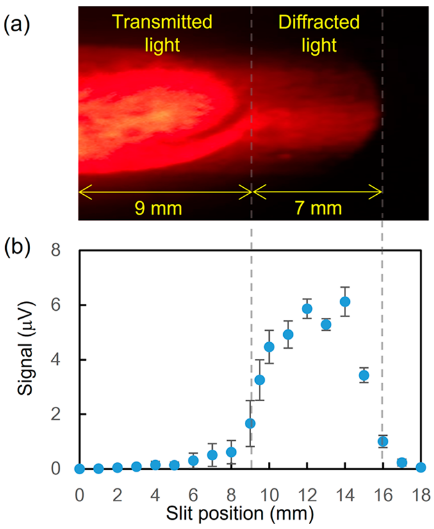

Figure 4. Dependences of signals on a detection position: (a) the probe beam image and (b) the signal intensity dependence on the slit position.

shows the signal intensity for each detection position. Clear signals were observed at the diffracted light region, while no or quite a low signal was observed at the transmitted light region. In addition, we confirmed that signals were not generated when either of the probe laser or the excitation laser was turned off. Also, signals were not observed for a blank solution (water) even at the diffracted light region. From these results, we concluded that the observed signal was derived from a change of the diffracted light intensity by the photothermal effect of target molecules.

Finally, we investigated the difference of the signal generation mechanism in the POD in comparison with that in TLM and photothermal deflection spectroscopy (PDS). For this, the signal intensity and phase were measured by changing a focal point of the probe beam and fixing a focal point of the excitation beam. In the TLM, the signal and phase depends on the difference of the focal points and the dependency is used as a proof of the detection principle.[46,47] Figure 5a shows an experimental arrangement. The channel position was set as the origin of the z-axis. The focal point of the excitation beam was fixed at the channel position to maximize the photothermal effect by target molecules in the channel.

The focal point of the probe beam (Δz) was moved along the z-axis by adjusting a beam expander, which was confirmed by the CCD camera and a scale of the z-axis stage. The results are shown in Figure 5b. Maximal signal intensity was obtained

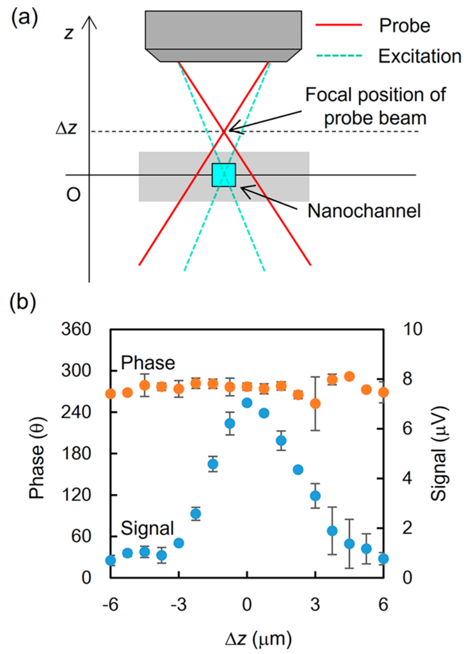

Figure 5. Dependences of the signal and phase on the focal position of the probe beam (Δz). (a) Schematic illustration of the experimental arrangement. The channel position is set at Δz = 0 and (b) the results.

at Δz = 0, and the signal intensity decreased as the Δz became larger. This result is in good agreement with the proposed detection principle of the POD, considering that maximal diffracted light intensity is obtained when the probe beam is focused on the channel. If the TLM and PDS are dominant, the maximum signal intensity should be observed at $\Delta z = \pm z_c$ (z_c, confocal length). In addition, no phase shift was observed around Δz = 0. The phase is related to the sign of signal for the modulated excitation beam, and the result shows that the diffracted light intensity change by the photothermal effect always increases or decreases. This result is also in good agreement with the detection principle of the POD, because the diffracted light intensity always increases by the photothermal effect, whereas the intensity change becomes reversed at Δz = 0 for the TLM and the PDS. Based on these investigations, the detection principle of the POD was confirmed.

Performance Evaluation. Figure 6 shows the calibration curve for the Sunset Yellow aqueous solution in a 400 nm wide × 400 nm deep channel. The concentration was determined on the order of 10^{-5} M. The calibration curve showed good linearity with the concentration. The LOD was defined as a concentration that gives a signal equivalent to blank + 3σ, and σ is calculated from the signal fluctuation of the sample at the lowest concentration. The calculated LOD was 5.0 μM, which corresponds to 500 molecules, considering a detection volume of 0.23 fL. The detection volume was determined from the excitation beam spot diameter (1.4 μm) and the channel width and depth (400 nm). This result shows that the POD has realized an ultrasensitive label-free concentration determination of 10^2 molecules in a single 10^2 nm (both of width and depth) channel for the first time. Also, the LOD is 1 order of magnitude lower than that of the DIC-TLM channel previously reported.[38] No other absorption-based detection

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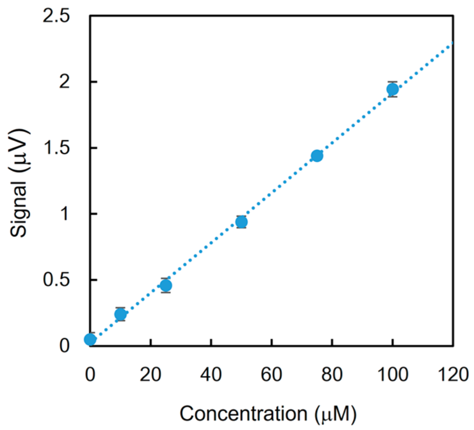

Figure 6. Calibration curve for Sunset Yellow aqueous solutions in a 400 nm wide and deep channel.

methods have realized such a sensitive detection of non-fluorescent molecules at the femtoliter scale volume.

Figure 7 shows a calibration curve for the Sunset Yellow aqueous solutions in nanochannels with five different sizes ranging from 200 nm wide × 200 nm deep to 1200 nm wide × 1200 nm deep. The power of the excitation beam was attenuated to 8 mW to avoid photodecomposition. The focal points of the two beams were adjusted to the center of each channel, and the slit position was optimized for each channel size to maximize the signal intensity. Each calibration curve showed good linearity, and the sensitivity (the slope of the calibration curve) decreased as the channel size became smaller. The sensitivity decrease is due to the detection volume decrease: a decrease in the number of target molecules in the excitation beam spot. To confirm this, the sensitivity per detection volume was calculated. Figure 8a shows the sensitivity per detection volume plotted as a function of the channel size. The sensitivity per detection volume for the DIC-TLM at the same experimental conditions was also plotted for comparison. In the case of the DIC-TLM, the sensitivity per detection volume largely decreased for the smaller channel, which was due to the thermal loss and the cancelation effect. However, in the POD, the sensitivity per detection volume did not show a significant change even for the 200 nm channel. A small decrease of sensitivity per detection volume for the smaller channel was explained by the diffracted light intensity decrease. Figure 8b shows the LOD for each channel size, showing no degradation of the LOD for the 200 nm channel. From these results, we concluded that POD was applicable to the 10^2 nm channel. The size limit of the nanochannel is under investigation.

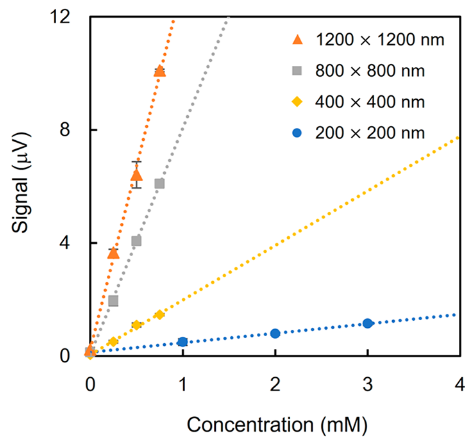

Figure 7. Calibration curves for nanochannels with the four different sizes.

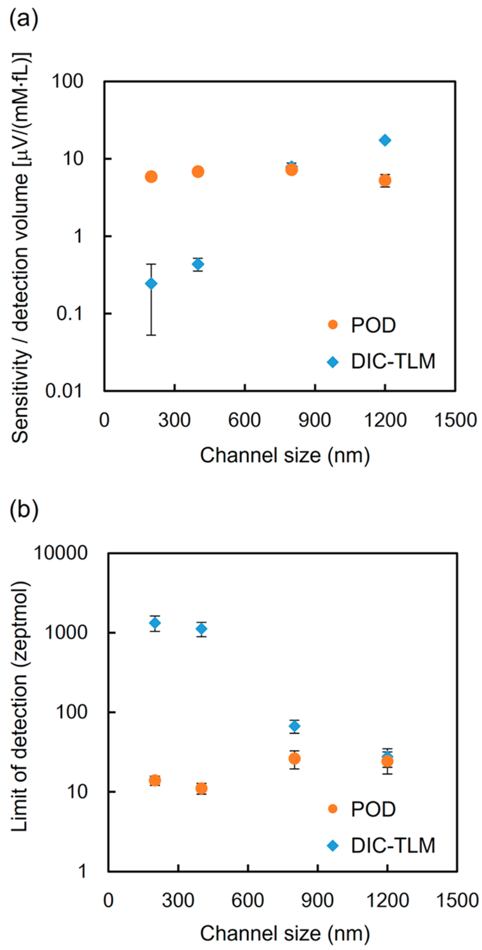

Figure 8. Comparison of the POD and the DIC-TLM: (a) sensitivity per molecule and (b) LOD.

■ CONCLUSION

A new detection principle, POD, was developed and realized for ultrasensitive and label-free molecule detection in 10^2 nm channels with an LOD of 500 molecules. The advantage of the POD is that detection of nonfluorescent molecules in ultrasmall sample volumes is possible without fabrication of additional nanostructures other than the nanochannel. Therefore, it will be used for analytical devices such as femtoliter–attoliter immunoassays and chromatography in 10^2–10^3 nm channels. By using UV laser as an excitation beam, biomolecules such as DNAs and proteins can be detected without labeling. For single-molecule analysis, 10^1–10^2 nm spaces are suitable because the ultrasmall volume enables precise manipulation of a single molecule and the POD will be used. Furthermore, the POD is applicable to the basic research of liquid properties and chemical reactions in the nanochannels. Thus, the POD will be an important detection tool for nanofluidics.

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K.M. conceived the project and designed the experiments. Y.T. designed and performed the experiments.

Notes

The authors declare no competing financial interest.

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